Manual

Special Functions for Line Production

MDE-SFL 8.1

Version 1.1.4716

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Contents

1 Overview – Special Functions for Line Production

Purpose

This component provides functions at the shop floor terminal for connection of lines.

The function package is used when you wish to connect a simple production line whose order-related consideration covers only one operation.

Integration

The component offers the connection of lines and their aggregates via corresponding machine interfaces for monitoring and for automatic recording of quantities.

Features

- Aggregate configuration
  Configuration possibility for aggregates and their assignment to lines
- Line and aggregate monitoring
  Monitoring of the line and its aggregates via machine interfaces and automatic posting of the recorded statuses.
- Quantity recording
  Order-related quantity recording and posting for the line
- Icon view at Windows terminals
  Special icon view at Windows-based line terminals

2 Line Assignment

Summary

Menu	Master data → Workplaces/ machines → Line assignment
Transaction code	mla
Function authorization	mdmla

The term flow manufacturing is used to describe a time-bound work process during which larger quantities of the same type of products are manufactured. The machines required in this process (also referred to as aggregates) are arranged by work sequence; the workpieces are usually transported automatically on a conveyor belt to the next aggregate. The consolidation of the separate aggregates is generally also referred to as a line.

Usage

Both aggregates as well as the line as the logical consolidation are defined as HYDRA machines. They are specifically identified and therefore set apart from "traditional" machines by the *machine type* identifier defined in the *Machine/ workplace configuration.*

Machine type	**Designation**
A	Aggregate
L	Line

The aggregates are assigned to a line using the aggregate assignment function in *Machine/ workplace configuration*. A selection dialog *Aggregate line assignment* appears, in which the aggregates assigned to the line are displayed.

Selection criteria

The application provides the following selection criteria:

Line

Used to select the line

Field descriptions

Line assignment

Line

Unique name for the line

Position

Position of the line at the terminal or on the display

Aggregate

The aggregate that the line is assigned to.

Workplace master data

Short designation

Short name for the aggregate (workplace/ machine)

Designation

Full name for the aggregate (workplace/ machine)

Group

Associated group for the aggregate (workplace/ machine)

Cost center

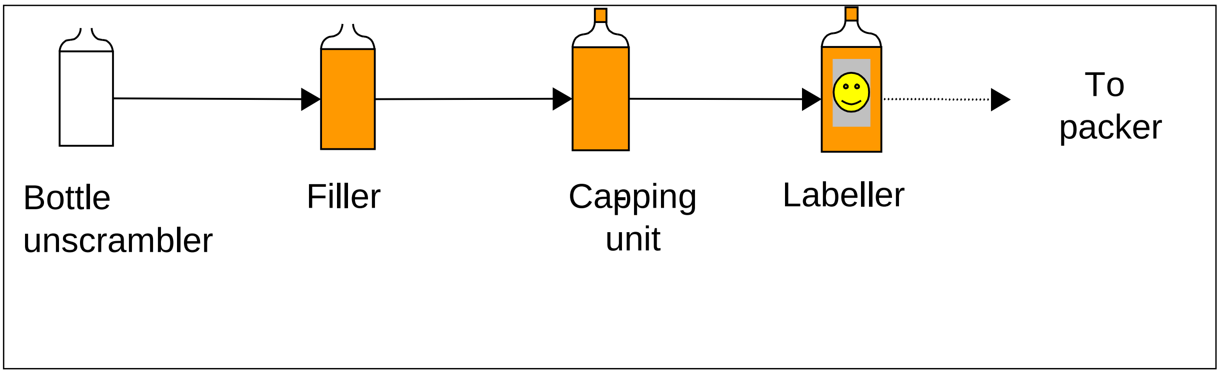
Associated cost center for the aggregate (workplace/ machine)

3 Configuration and Function of the Specific Line Processing

Overview

The term flow manufacturing is used to describe a time-bound work process during which larger quantities of the same type of products are manufactured. The machines required in this process (also referred to as *aggregates*) are arranged by work sequence; the work pieces are usually transported automatically on a conveyor belt to the next aggregate. The consolidation of the separate aggregates is generally also referred to as a *line*.

The following chart illustrates and example of the production process:



Creation of Line and Aggregates

Both the aggregates and the line as a logical consolidation are defined in the system as MNR Type resources. They are specifically identified and therefore set apart from "traditional" machines by the identifier *workplace type* in the resource configuration:

Workplace type	Designation
A	Aggregate
L	Line

Aggregate Assignment

The aggregates are assigned to a line. The Line assignment configuration is provided for this.

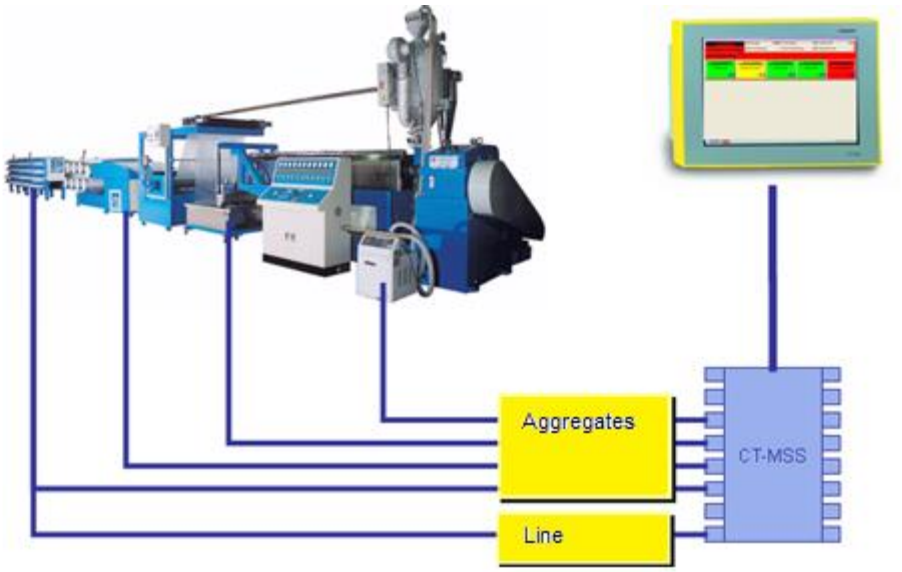
Machine Connection of Aggregates and Line

After selection of a line or aggregate, the settings for the machine connection are made per line/aggregate under the tab *MDE configuration* in the resource configuration. The following points must be observed:

Connection of the Line

The line is monitored by means of a direct machine connection. The signal there (cycle signal or operating signal) comes from an aggregate in the line which is regarded as being representative for the line (e.g. bottler). A separate contact has to be wired on the shop floor unit (e.g. CT-MSS, CT-UMPS) for this.

For order-related quantity determination, the line must be wired via corresponding counter inputs. In the counter configuration, the corresponding counter inputs then have to be defined via which the cycles are to be transmitted from an aggregate to the terminal. If these cycles are also to be posted machine-specifically to the aggregate, a separate contact has to be wired on the shop floor unit (e.g. CT-MSS, CT-UMPS) for this aggregate.



Connection of the Aggregates

By analogy with the line, an individual aggregate is also monitored by means of a direct machine connection. The signal there (cycle signal or operating signal) comes from the respective physical aggregate. Here again, separate contacts have to be wired on the shop floor unit for each aggregate.

Note on field *Cycle extension (*tab *MDE configuration → Monitoring)*:

> The ERP system defines a cycle specification (target duration for 1000 machine cycles) for each operation. This value is multiplied by a tolerance value (in percent, defined for each machine/aggregate) and results in the target cycle for the cycle monitoring at the terminal.

As only one cycle specification can be defined for the operation, this specification influences the cycle monitoring for all the aggregates assigned to the line (cf. *Aggregate assignment*). It is therefore important that a tolerance value is entered for the aggregates which, based on the cycle specification, results in a realistic target cycle for the cycle monitoring.

Assignment Machine Status → Aggregate or Line

The settings for the machine connection are made per line/aggregate via the *Status assignment*. A precondition for the status assignment is the creation of the status texts.

Machine Status of the Line

At least the following statuses have to be defined for the line:

- **Status** — ***Production***
  - Status number — freely variable, e.g. 1
  - Status text — freely variable, e.g. PRODUCTION
  - RPA — MUT
  - Production identifier — Production
  - Transmission to aggregates — No
  - Manual assignment — No
  - Automatic assignment — Yes, if monitoring is performed via operating signal
  - Assignment number — Input at which the operating signal *Production* is received

  Further settings are optional.
- **Status** — **Disturb**ance
  - Status number — freely variable, e.g. 99
  - Status text — freely variable, e.g. GEN. DISTURBANCE
  - RPA — freely variable, generally: DCI
  - Production identifier — Gen. disturbance
  - Transmission to aggregates — No
  - Manual assignment — No
  - Automatic assignment — No
  - Assignment number — 0

  Further settings are optional.

If further order-relevant statuses are to be recorded, these are also defined for the line. The following example shows the configuration taking the example of the status *Setup*.

- **Status** ***Setup***

Status number	freely variable, e.g. 2
Status text	freely variable, e.g. SETUP
RPA	freely variable, SET
Production identifier	Disturbance
Transmission to aggregates	Optional, e.g. for setup: Yes
Manual assignment	Yes
Automatic assignment	No
Assignment number	0

Further settings are optional.

Note for Option *Transmission to Aggregates:*

The option *Transmission to aggregates* offers the possibility of setting a global status on the line which is then set automatically for all aggregates assigned to that line. This eliminates the need for explicit setting of this status on each individual aggregate, e.g. during setup.

Note here that this status is also defined for all aggregates, and that the status number has to be identical with the status number of the line.

Example:

Line S:	Status Setup	Status number: 2
Aggregate S-A1:	Status Setup	Status number: 2
:	:	:
Aggregate S-A5:	Status Setup	Status number: 2

Machine Status of the Aggregates

By analogy with the line, at least the statuses for Production and for Gen. disturbance also have to be defined for the aggregates. As an option, further aggregate-related status can be defined (e.g. Electrical fault).

If the statuses were defined on the line as *Transmission to aggregates*, these have to be additionally defined for each aggregate. Note here that no production lock may be activated with these statuses, as this lock would continue to be activated at the terminal even when the status is changed and would have to be deactivated manually.

Settings at the Terminal

In order that the status of the line and its aggregates can be optimally monitored at the terminal, a display in icon view is recommended. For this the *Terminal configuration* has to be called up. After selection of the corresponding terminals and selecting the tab *MF functions*, select the option *Symbols* under *Other settings*.

All other terminal settings are independent of the line settings.

Assignment of the Line to the Terminal

Both the line and its aggregates are assigned to the terminal. During the assignment of a line to the terminal, the associated aggregates are automatically also assigned and are displayed at the end of the list with position 99.

It is possible to assign several lines to one terminal (Windows terminals only): With CT83x max. 2 lines, with CT84x max. 3 lines. An assignment of both lines and individual machines (e.g. packing unit) to a common terminal is also possible. An assignment of lines, aggregates or machines to several terminals, however, is not possible.

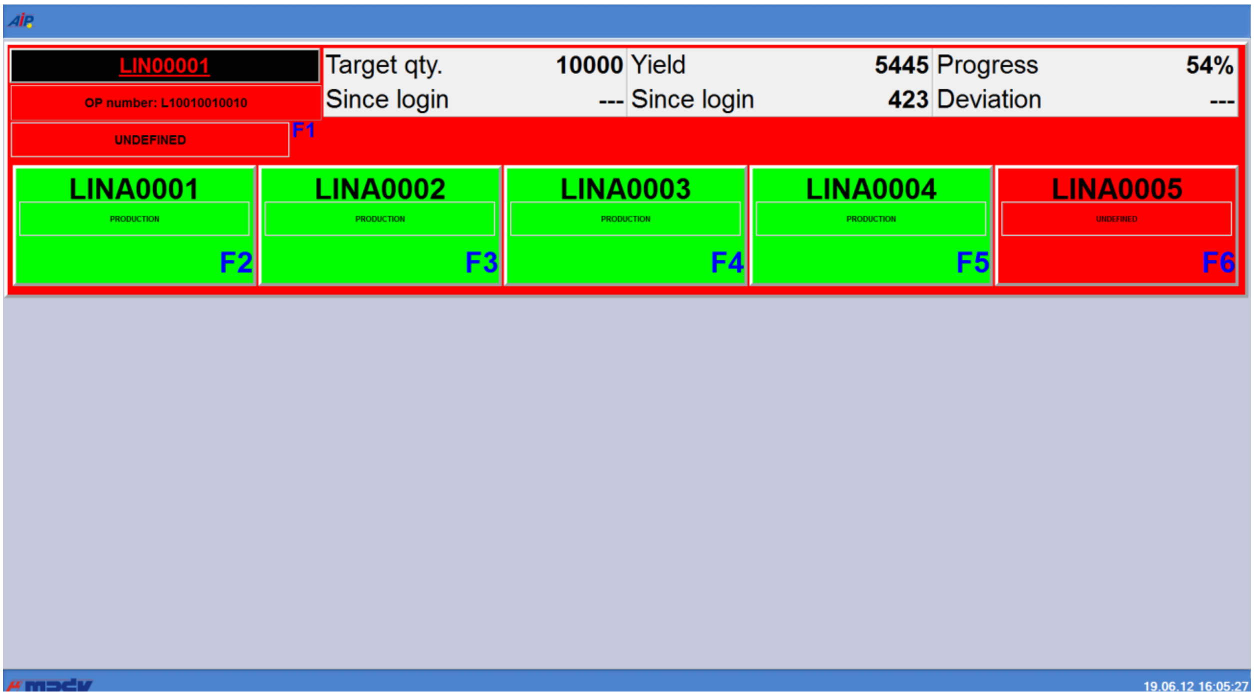
Please observe the limit of 16 HYDRA machines (lines + aggregates) per terminal.

4 Handling at the Terminal

For the display and handling of requirements which occur during flow manufacturing (lines), the dialogs and functions described below are modified at the terminal.

Layout

If Machines/workplaces *Symbols was* selected as display during the terminal configuration, the lines and its aggregates are displayed as follows:



Each line is displayed as a box across the whole screen width. Its assigned aggregates are displayed below this box. The sequence in the display thereby corresponds to the sequence of the assignment of the aggregates to the line.

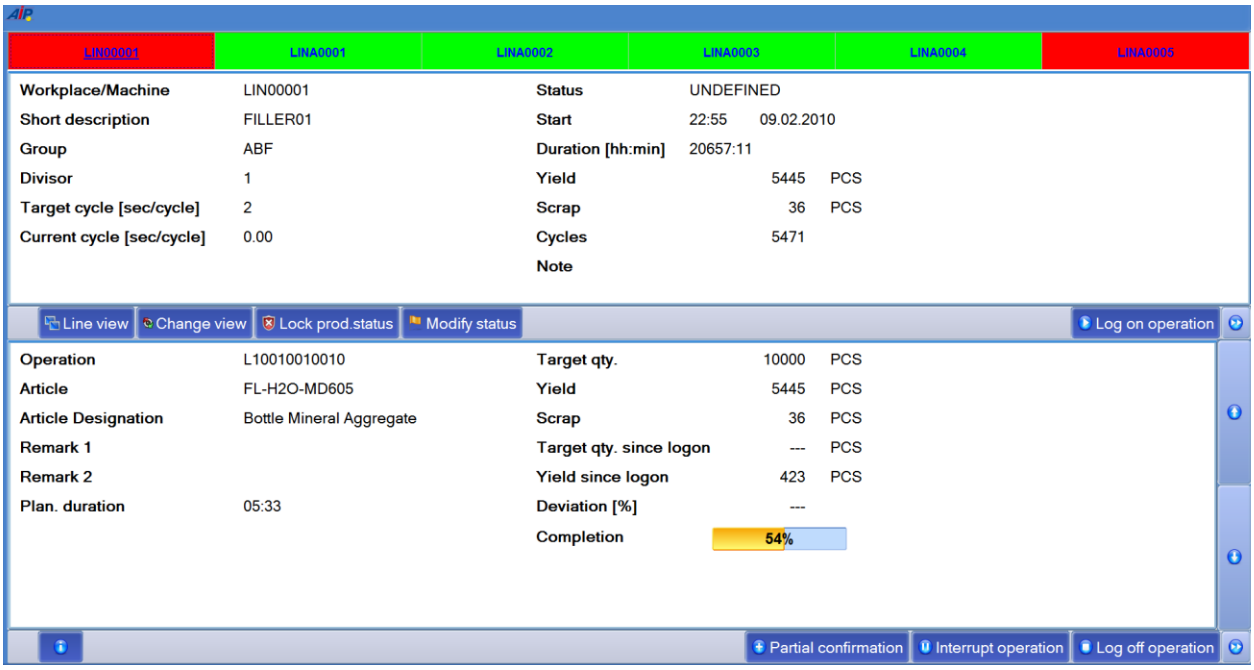
If further lines or machines are assigned to the terminal, these are displayed underneath.

The color in which the line, its aggregates and/or further machines is displayed corresponds to the momentary status:

- Production green
- Disturbance yellow
- Not assigned red

If further machines are assigned, these are displayed below the lines. The width of the display depends on the number of machines to be displayed.

Touching the corresponding icon (*touchscreen*) or pressing the assigned key (is displayed in the bottom right corner of each icon) switches the display from the icon view to the machine overview.



Touching the button ESC returns you to the icon view again.

If no further inputs are made, the view changes automatically after 30 seconds to the icon view again.

Postings at the Terminal

Depending on the posting to be carried out, the line, the corresponding aggregate or the desired machine first has to be selected. The current selection is indicated by the box around the designation and by the display in the upper part of the *Machine overview*.

Order-related or personalized postings can only be made for a line or a machine.

Status changes, on the other hand, can be made on a line, a machine or an aggregate.

Status Change on the Line

If machine statuses are defined for a line for which the switch *Transmission to aggregates* has been set, a manual status change on the line (e.g. Setup) results in a status change of all the aggregates assigned to the line (in this case also to the status Setup). It makes no difference here whether the status was changed with the function "Change status" or by entering the status in the operation postings (e.g. operation interruption, operation logoff). A precondition, however, is that the status number at the respective aggregates is identical with that on the line.

The procedure at the terminal is as follows:

When the status for the line has been set, it is automatically assigned to the aggregates assigned to the line (a small window appears in which the progress of the status transmission to the aggregates is displayed).

Production Lock

Touching the button "Lock production status" allows the production lock to be activated and deactivated. It is thus possible to prevent switching to the status *Production* despite incoming machine pulses (e.g. during Setup).

If this function is set for a line, it automatically applies to all the aggregates in the line. In reverse, when the production lock is deactivated on the line, it is also deactivated on the assigned aggregates.

Please note

If the production lock was set for the line, it can nevertheless be deactivated individually, i.e. for an individual aggregate

During the transmission of the production lock from the line to the aggregates, the event P_SPERRE (production lock) is not logged for the aggregates. The setting of the production lock is consequently documented in the machine history only for the line.

Partitioning

As no operations are logged on at the aggregates, only the machine-specific partitioning and the machine-specific pulse factor are taken into consideration for the aggregates when determining the quantities. Order-specific partitioning or pulse factor play no role at aggregates.

A change of partitioning which is only permitted on the line leads to a change in the order-related partitioning. The machine-related partitioning can only ever be changed in the machine configuration.

Target Cycle

The ERP system defines a cycle specification (target cycle) for each operation. This value is multiplied by a tolerance value (in percent, defined for each machine/aggregate) and results in the target cycle for the cycle monitoring at the terminal. As only one cycle specification can be defined for the operation, this specification applies to all the aggregates of a line on which the operation is logged on.

With the corresponding authorization it is possible at the terminal to change the target cycle for the operation (e.g. if it is discovered that the cycle specification set by the ERP system for the operation is too small). Changing of this target cycle influences the cycle monitoring at all the aggregates. It is therefore important that a tolerance value is entered for the aggregates which, based on the target cycle, results in a realistic target cycle for the cycle monitoring.